

Quantum Machine Learning for Near-Term Feasibility: A Survey of Emerging Directions and Challenges

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We conducted a large-scale survey of Quantum Machine Learning (QML)-related preprints from arXiv to investigate potential, yet elusive, advantages of QML in real-world scenarios. To identify promising near-term application areas and representative use cases for QML, we systematically analyzed publication and citation trends across preprints. Our analysis identified *Medicine and Drug Discovery*, *Chemistry*, *Finance and Fraud Detection*, and *Physics* as primary application areas, with *Medical Imaging Diagnosis* and *Credit Card Fraud Detection* emerging as particularly notable use cases. Addressing the expected advantages and practical challenges of QML, we examined empirical validations specifically within these two use cases. Encouraging results were observed for each use case: identifying subtle patterns in neuroimaging data and performing unsupervised anomaly detection. However, definitive evidence remains inconclusive, highlighting significant challenges related to current quantum hardware limitations, learning difficulties such as Barren Plateaus and exponential concentration phenomena, and computational overhead.

I. INTRODUCTION

Quantum Machine Learning (QML) has attracted attention as a promising domain for quantum computing. In QML, there are two primary approaches: one involves accelerating specific parts of classical algorithms (such as linear algebra and optimization tasks) using quantum algorithms [1], while the other leverages quantum properties to develop novel machine learning algorithms that surpass classical algorithms [2–5]. The former approach typically requires Fault-Tolerant Quantum Computing (FTQC), making it a long-term prospect. Thus, this survey specifically adopts the latter approach, emphasizing near-term feasibility, particularly those realizable with Noisy Intermediate-Scale Quantum (NISQ) computers [6–8].

In QML, algorithms leverage the high-dimensional information capacity of quantum Hilbert spaces and quantum mechanical properties such as superposition, entanglement, and interference. These properties are expected to enable the processing of high-dimensional data and the recognition of complex patterns, tasks that are challenging for Classical Machine Learning (CML) [2–5]. Moreover, from the industrial perspective, QML is regarded as a promising candidate with substantial potential for economic impact in areas such as automotive, finance, search and advertising, as well as other sectors [9].

Despite these high expectations, the advantages of QML over CML are still not fully understood. This is primarily because, due to limitations in quantum hardware, demonstrating quantum advantages in QML predominantly relies on theoretical approaches, while CML is mainly developed through empirical successes whose theoretical explanations remain incomplete [10]. Currently demonstrated quantum advantages in QML apply only to highly specific problems [11, 12]. Although these results serve as supportive evidence for QML, its advantages in practical, real-world use cases remain unclear.

Our objective in this survey is to bridge this gap, specifically focusing on applications and use cases rather than theoretical aspects. This survey is structured around two major research questions. First, we investigate which areas and use cases are promising for QML in near-term applications. Second, we investigate what advantages are expected and what challenges exist regarding the practical realization of QML.

To address the first research question, we conducted a large-scale survey of the arXiv preprint repository. We analyzed abstract texts related to QML through statistical and Natural Language Processing (NLP) techniques to identify the main QML research topics. By assessing the relative prominence of these topics based on publication volume and citations, we gained insights into promising areas and use cases receiving heightened attention. As mentioned at the outset, this survey specifically targets research examples designed for NISQ computers, rather than those targeting FTQC. We also focus specif-

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ically on gate-based quantum computing due to its universality and flexibility, which aligns with a broad range of machine learning tasks.

To address the second research question, we further conducted an in-depth investigation into the empirical validations performed for use cases identified as promising in the earlier survey. Specifically, we examined the current state, challenges, and expected advantages of QML algorithms for practical implementation. From the perspective of quantum advantage, we adopted the three criteria proposed in prior work [13], assessing whether QML surpasses CML in terms of accuracy, speed, and energy consumption.

Through this survey, we identified four promising application areas: (1) *Medicine and Drug Discovery*, (2) *Chemistry*, (3) *Finance and Fraud Detection*, and (4) *Physics*. Focusing on two of these areas, *Medicine and Drug Discovery* and *Finance and Fraud Detection*, we also identified three promising use cases: (1) *Drug Design and Evaluation*, (2) *Medical Imaging Diagnosis*, and (3) *Credit Card Fraud Detection*. We then conducted detailed investigations of two of these use cases, *Medical Imaging Diagnosis* and *Credit Card Fraud Detection*. These investigations clarify the current status, challenges, and future directions regarding the expected quantum advantages of QML.

While recognizing the limitations inherent to our study, such as constrained survey scope and ongoing lack of definitive evidence for quantum advantages, we aim to provide useful insights to support researchers and practitioners in evaluating future opportunities and challenges within the QML domain.

The remainder of this paper is organized as follows. Sections II–IV present our survey addressing the first research question. In Section II, we systematically extract and organize QML research topics, briefly detailing the statistical methods used, followed by presenting the resulting topic structure derived from arXiv data. In Section III, we analyze trends associated with the extracted topics, considering both overall publication volume and citation-based metrics. We then select key topics based on these analyses for further exploration. In Section IV, we identify and categorize use cases derived from selected key topics, further highlighting those receiving attention based on publication volume and citations. Sections V and VI present our survey addressing the second research question. In Section V, we evaluate the experimental validations, reported outcomes, and potential quantum advantages of these selected use cases. In Section VI, we suggest specific validation approaches necessary for further investigating the surveyed use cases and their potential quantum advantages. Finally, Section VII summarizes our findings and suggests future research directions.

II. SYSTEMATIC EXTRACTION OF QML RESEARCH TOPICS

In this section, we present a methodological framework for systematically identifying and structuring research topics in QML, drawing on a large-scale survey of preprints from the arXiv repository. To this end, we collected abstract texts from 2539 preprints categorized under “quant-ph” and associated with the keyword “machine learning” [14]. All preprints were published by the end of September 2024, with the earliest dating back to November 2004. Citation data were retrieved from Semantic Scholar [15] on October 2, 2024.

The topic extraction process involved two complementary approaches. The first methodology employed statistical methods and NLP techniques, namely, term frequency-inverse document frequency (tf-idf) [16, 17] and t-distributed stochastic neighbor embedding (t-SNE) [18], to the entire dataset. Abstract texts were vectorized and projected onto a two-dimensional plane, where clusters were identified using Kernel Density Estimation (KDE) [19, 20]. We then manually labeled these clusters based on frequently occurring keywords, resulting in what we refer to as “statistically extracted topics.”

The second approach targeted the top 10% (245 preprints) with the highest number of citations. Each of these preprints then underwent expert inspection to identify both application and technical topics. Here, application topics refer to specific industrial domains or use cases, whereas technical topics encompass QML methodologies such as model architectures and performance evaluations.

As a final step, we merge and systematically organize the topics obtained from both approaches. We note that expert inspection generally yields more comprehensive coverage, whereas the statistically extracted topics are less exhaustive. Nevertheless, one particular topic emerges solely from the statistical analysis.

Below, we first offer a detailed explanation of the two approaches discussed. We then present a comprehensive list and categorization of the extracted topics.

A. Statistical Topic Extraction Using NLP and Dimensionality Reduction

In this subsection, we outline our statistical and NLP-based approach for topic extraction.

The procedure began with preprocessing the abstract texts of the collected preprints from arXiv. Preprocessing steps included converting all text to lowercase, removing symbols, stemming words, and discarding single-character tokens. Additionally, any word appearing in over 30% of the preprints or in fewer than three preprints was removed to reduce noise and maintain feature relevance.

After preprocessing, we vectorized the abstract texts using the tf-idf method via scikit-learn’s

TfidfVectorizer [21]. We then applied the t-SNE algorithm (also via scikit-learn’s TSNE) to reduce dimensionality and project the high-dimensional data onto a two-dimensional plane (see Figure 1).

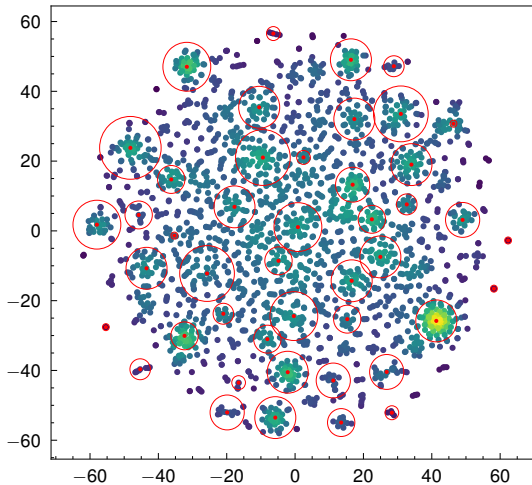


FIG. 1. Two-dimensional projection of arXiv preprints obtained via tf-idf vectorization and t-SNE-based dimensionality reduction. The preprint density, computed by KDE, is shown using a color scale from dark purple (low density) to bright yellow (high density). Red points mark local density maxima, while red circles indicate cluster boundaries determined by incremental radius expansion.

This projection naturally groups preprints with similar abstract texts, leading closely related research to appear near each other on the plane, thereby forming distinct clusters. To systematically identify these clusters, we employed KDE via SciPy’s `gaussian_kde` [22] to measure the density at each preprint’s location. Each preprint was assigned a density value computed by KDE at its respective location. Points on the plane without any corresponding preprints were excluded from further consideration.

Clusters were identified by first locating local density maxima (red points in Figure 1). Starting from these points, the cluster boundaries were expanded incrementally. If any newly included preprints had lower density values than those already in the cluster, the expansion continued; conversely, if their density values were higher or equal, the expansion was halted, finalizing the cluster boundary. This approach ensured that each cluster encompassed preprints in descending density regions radiating from the local maximum.

Finally, we manually examined the top keywords in each cluster to assign thematic labels. These resulting clusters and labels, referred to as “statistically extracted topics,” are presented in detail in Appendix A.

B. Expert Inspection-Based Topic Extraction from Highly-Cited Preprints

In this subsection, we describe the expert inspection approach used to extract topics from highly-cited preprints.

We began by grouping 2539 arXiv preprints by publication year and ranking them according to their number of citations to determine the top 10% threshold for each year. For years prior to 2013, when relatively few preprints were published, we did not calculate thresholds initially but instead established them later via extrapolation. Figure 2 (left panel) shows each preprint plotted by publication date and citation count, with orange points indicating the 10% threshold.

Next, we defined daily thresholds by linearly interpolating the annual thresholds, assuming each year’s threshold applied on July 1st. Because an unusually high number of citations was observed for 2018, we adjusted that year’s threshold by averaging the thresholds of 2017 and 2019. We extrapolated threshold values prior to July 2017 using the threshold on July 1, 2017, and likewise extrapolated values after July 2024 using the threshold on July 1, 2024. We then selected preprints whose number of citations exceeded their respective daily thresholds, yielding 245 highly-cited preprints (see Figure 2, right panel). Table I summarizes the number of preprints classified as highly-cited each year.

All 245 extracted preprints then underwent an expert inspection, during which each abstract text was examined to identify relevant topics from both application and technical perspectives. We classified as application topics those that relate to specific industrial domains or use cases. However, purely benchmark-oriented examples, such as MNIST digit recognition, were excluded from the application topics. We also classified as technical topics those that refer to methodologies, model architectures, and performance evaluations.

As a result, the expert inspection approach accounted for nearly all topics covered in this survey. Topics identified through the statistical extraction approach were largely a subset of those from the expert inspection approach, with the sole exception being the application topic *Logistics*. A complete list of these topics is presented in the following subsection (see Tables II–III).

C. Merged and Organized List of Extracted Topics

In this subsection, we present the integrated set of topics obtained from both statistical extraction and expert inspection.

As described in the previous subsection, our topic organization primarily relies on the expert inspection approach due to its comprehensive coverage. In addition, we included the application topic *Logistics*, identified exclusively through the statistical extraction approach.

Table II summarizes the application topics identified

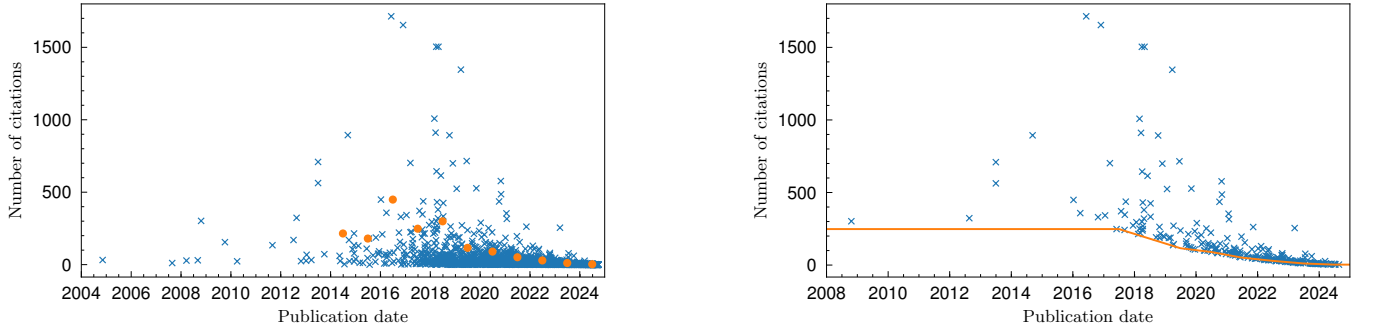


FIG. 2. Left panel: Scatter plot of preprints by publication date (horizontal axis) and number of citations (vertical axis), with orange points marking the top 10% threshold for each year. Right panel: Scatter plot focusing on the 245 highly-cited preprints, where the daily interpolated or extrapolated thresholds are shown as an orange line.

TABLE I. Number of collected preprints per year, number identified as highly-cited, and corresponding proportion selected.

Year	Preprints	Highly-cited	Proportion Selected
2004	1	0	0%
2005	0	0	0%
2006	0	0	0%
2007	1	0	0%
2008	3	1	33%
2009	1	0	0%
2010	1	0	0%
2011	1	0	0%
2012	3	1	33%
2013	6	2	33%
2014	11	1	9%
2015	14	0	0%
2016	28	5	18%
2017	67	8	12%
2018	139	27	19%
2019	183	19	10%
2020	264	29	11%
2021	353	35	10%
2022	457	48	11%
2023	538	52	10%
2024*	468	17	4%
Total	2539	245	10%

*As of September 2024.

in our analysis, named according to their relevant industry sectors. Because the number of preprints available for expert inspection is limited, further subdivision is not feasible at this stage. A more detailed discussion of each application topic will appear in subsequent sections. Although the “Preprints” column in Table II reflects counts only from the 245 highly-cited preprints, we will re-examine these counts using the full set of 2539

preprints in the following section.

We also provide a list of technical topics in Table III, which are broadly grouped into six categories: *Model*, *Data*, *Performance*, *Quantum-Classical Synergy*, *Privacy Enhancement*, and *Quantum Speedup*. We use *Quantum-Classical Synergy* to refer to methods that combine quantum and classical approaches beyond standard hybrid schemes. A brief overview of each technical topic is given in Appendix B.

III. TREND ANALYSIS OF TOPICS BY PUBLICATION VOLUMES AND CITATIONS

In this section, we present a trend analysis of QML topics by examining their publication volumes and citations, based on the topics identified in the previous section. The main objective is to systematically identify key topics that have gained increased attention in recent years. Since our focus is on potential industrial applications, we specifically emphasize application topics as candidates for key topics, while treating technical topics in a supplementary manner.

To achieve this, we first conducted a keyword search on a total of 2539 arXiv preprints that had been previously collected. We defined sets of keywords to identify each application and technical topic. Preprints containing at least one instance of the predetermined topic-related keywords in their abstract texts were flagged as covering the corresponding topic. Consequently, a single preprint could be associated with multiple topics, both application and technical.

Subsequently, we aggregated the preprints associated with each application topic [53]. We then calculated several metrics to quantify the relative attention these topics received in terms of publication volume, citations, and publication trends.

By comparing these metrics across different application topics, we identify key topics that demonstrate increased research activity and growing prominence in recent years. In addition, to facilitate more targeted exploration in subsequent analyses, we also tabulate the

TABLE II. List of identified application topics, along with corresponding use case summaries and the number of associated preprints. The “Preprints” column shows the frequency of each topic within the 245 highly-cited preprints subjected to expert inspection. Because individual preprints may cover multiple application topics, the total counts represent cumulative occurrences rather than unique preprints.

Application Topic	Summary of Corresponding Use Case(s)	Preprints
Physics	Event detection / classification in high-energy physics experiments [23–28] Supernova / Stellar classification [29, 30]	8
Medicine and Drug Discovery	Medical image diagnosis [31] Drug design and evaluation [32–35] Comprehensive review of this topic [36, 37]	7
Finance and Fraud Detection	Derivative hedging optimization [38] Credit card fraud detection [39, 40] Comprehensive review of this topic [41–44]	7
Chemistry	Toxicity prediction from molecular graph structures [45] Prediction of energies and force fields for molecular dimers [46] Comprehensive review of this topic [33]	3
Aerospace	Mission planning for earth observation satellites [47]	1
Disaster Prevention	Forecasting future river water levels and flood occurrences [48]	1
Fluid Dynamics	Surrogate model for fluid dynamics simulations [49]	1
Recommendation	Recommendation based on user profiles and content information [50]	1
Logistics	Solving vehicle routing problems (VRP) [51, 52]	2*
Total		31

*The topic *Logistics* was identified exclusively through statistical topic extraction; therefore, the items for this topic are based on two preprints included in the associated cluster. As shown in Table XIV, this cluster includes three preprints, one of which was found to have been withdrawn and excluded from this count.

TABLE III. List of technical topics identified through expert inspection and statistical extraction, organized into six categories. Definitions and detailed explanations for each topic are provided in Appendix B. Because this survey does not focus on FTQC, the category *Quantum Speedup* is not subdivided further.

Category	Technical Topic(s)
Model	Variational methods: Parametrized Quantum Circuit, Quantum Neural Network*, Quantum Circuit Born Machine, Reinforcement Learning, State diagonalization, QAOA-clustering, Quantum Optimal Control
	Non-variational methods: Quantum Kernel, Quantum Reservoir Computing, Imaginary Time Evolution
Data	Quantum Embedding, Quantum Data, Time Series, Graph, Physics-informed Learning
Performance	Expressivity, Generalization, Trainability, Adversarial Attack
Quantum-Classical Synergy	Classical Surrogate/Shadow, Tensor Network, Training Classical Model
Privacy Enhancement	Circuit Encryption, Federated Learning, Quantum Differential Privacy
Quantum Speedup	Quantum Speedup

*Topics related to *Quantum Neural Network* are omitted here because of their volume. For a complete list, see Appendix B.

frequency of associated technical topics appearing in preprints identified under these key application areas.

In what follows, we first describe our keyword-based methodology for topic identification, then detail the statistical aggregation and comparison of application topics, and finally present the key topics selected for deeper examination.

A. Keyword-based Identification of Application and Technical Topics

In this subsection, we outline the methodology used to identify application and technical topics in each preprint through keyword searches.

Based on the expert inspection described in the previous section, we clarified the definitions and representative examples for each topic. Using this information, we compiled a set of keywords, synonyms, and related terms to facilitate the detection of these topics in abstract texts. When necessary, we also specified exclusion keywords. The keywords assigned to each application topic are listed in Table IV, while those for technical topics can be found in Appendix C.

We now detail the exceptional keyword settings used to exclude unintended matches when identifying application topics.

First, to avoid general overview publications, we explicitly designated “review,” “lecture,” “survey,” and “overview” as exclusion keywords for all application topics.

Second, for the topic *Physics*, an unrestricted keyword search would have yielded a large number of unintended matches since quantum computing is inherently part of the broader field of physics. Therefore, our keyword settings were configured to focus specifically on high-energy physics and astrophysics, as identified by expert inspection. Other subfields of physics were not included under this keyword-based approach.

Finally, for the topics *Medicine and Drug Discovery* and *Chemistry*, we anticipated a significant overlap due to their closely related scopes. To avoid this overlap, keywords defined for *Medicine and Drug Discovery* were simultaneously set as exclusion terms for *Chemistry*.

We then performed preprocessing on each abstract text by converting all text to lowercase and replacing ASCII punctuation and symbols with whitespace. Next, we collapsed multiple consecutive whitespace characters into a single space. Each preprint was flagged as containing a topic if it included at least one of the assigned keywords in the preprocessed abstract. However, if a preprint contained any of the designated exclusion keywords, the corresponding topic flag was removed.

B. Relative Attention Metrics for Application Topics

This subsection explains the metrics used to identify key topics that have recently gained increased attention. We then present the results of our analysis.

For each application topic, we aggregated preprints identified in the previous step and computed metrics such as the annual number of preprints, the total number of preprints over the entire study period, and the number of citations. To exclude preprints unrelated to QML, we removed any preprints whose abstract texts did not contain “quantum machine learning,” “quantum learning,” or “qml,” and did not match any of the technical topics. This filtering was intended to minimize the inclusion of preprints not aligned with the scope of this survey, such as studies employing CML techniques for analyzing quantum systems or performing error mitigation.

Figure 3 illustrates the total number of preprints (left panel) and total citations (right panel) for each application topic over the entire study period. *Medicine and Drug Discovery* and *Chemistry* prove to be the most prominent, ranking highly in both publication volume and citations. These two topics are followed by *Finance and Fraud Detection* and *Physics*, both displaying substantial scholarly interest compared to the other topics. Although the topic *Recommendation Systems* comprises relatively few preprints, it exhibits a high number of citations, primarily due to three influential papers [50, 54, 55]. However, those particular preprints are outside the scope of this survey because they focus on FTQC-targeting or quantum-inspired algorithms rather than NISQ algorithms. Thus, despite the high number of citation, the prominence of *Recommendation Systems* does not necessarily reflect its relevance to the main focus of this survey.

Additionally, we quantify publication trends over time. Figure 4 shows the time series of annual number of preprints for each topic (left panel) alongside the linear regression coefficients of the time series indicating publication growth rates (right panel). Since the number of preprints before 2015 is largely zero, those early years are excluded from the regression analysis and plots. The data for 2024, available only up to the end of September, are plotted with a dashed line but also excluded from the regression calculations. These results indicate significant upward trends for *Medicine and Drug Discovery*, *Chemistry*, *Finance and Fraud Detection*, and *Physics*. Consequently, these topics demonstrate the most rapidly increasing research activity in recent years.

C. Identification of Key Application Topics

Based on the previously calculated metrics, *Medicine and Drug Discovery* and *Chemistry* are most prominent application topics, receiving the highest levels of attention in terms of publication volume, citations, and publication trends. These are followed closely by *Finance and Fraud Detection* and *Physics*, which also exhibit significant prominence compared to other application topics. Accordingly, we identify these four application topics as key candidates for further investigation.

To facilitate more detailed analyses in subsequent sections, we count the frequency of technical topics in preprints associated with each candidate key application topic. The results are summarized in Tables V–VIII. See Appendix B for a description of each technical topic.

According to Table V, preprints related to *Medicine and Drug Discovery* most frequently include the technical topic *Transfer Learning*, followed by *Model*-category topics such as *Quantum Kernel* and *Quantum Neural Network*. *Data*-category topics like *Quantum Data* and *Quantum Embedding* also rank high. Although fewer in number, *Privacy Enhancement*-category topics (e.g., *Federated Learning*) and the topic *Adversarial Attack* are

TABLE IV. Keywords and exclusion keywords for application topics.

Application Topic	Keyword(s)	Exclusion Keyword(s)
Physics	“high energy physics,” “lhc,” “astrophys,” “supernova”	“review,” “lecture,” “survey,” “overview”
Medicine and Drug Discovery	“bio,” “drug,” “health,” “medicine,” “medical”	“review,” “lecture,” “survey,” “overview”
Finance and Fraud Detection	“finance,” “financial,” “fraud”	“review,” “lecture,” “survey,” “overview”
Chemistry	“chemistry,” “chemical”	“review,” “lecture,” “survey,” “overview,” “bio,” “drug,” “health,” “medicine,” “medical”
Aerospace	“aerospace,” “satellite”	“review,” “lecture,” “survey,” “overview”
Disaster Prevention	“disaster,” “climate change”	“review,” “lecture,” “survey,” “overview”
Fluid Dynamics	“fluid dynamics”	“review,” “lecture,” “survey,” “overview”
Recommendation	“recommendation system”	“review,” “lecture,” “survey,” “overview”
Logistics	“logistic,” “vehicle routing problem”	“review,” “lecture,” “survey,” “overview”

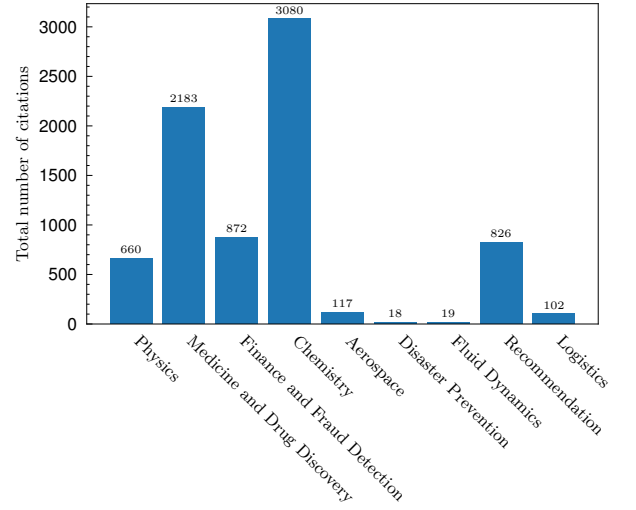
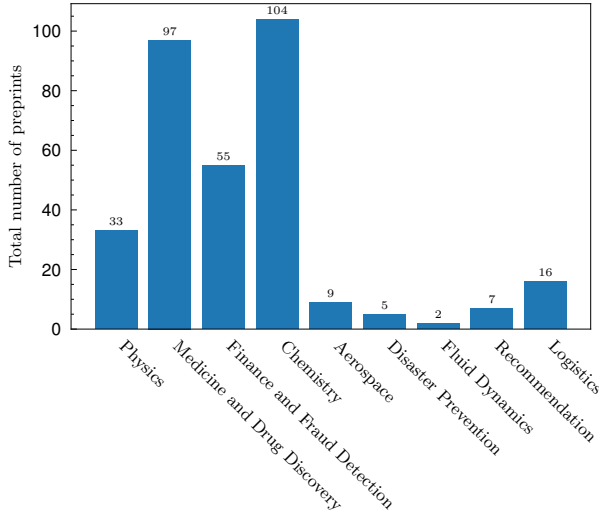


FIG. 3. Total number of preprints (left panel) and total number of citations (right panel) for each application topic across the entire study period.

notable, suggesting research directions that prioritize privacy and security concerns.

As shown in Table VI, preprints in *Chemistry* frequently address *Performance*-category topics such as *Trainability*, *Generalization*, and *Expressivity*, alongside *Data*-category topics like *Quantum Embedding* and *Quantum Data*. In the *Model*-category topics, *Quantum Neural Network* is also well represented.

Table VII indicates that preprints in *Finance and Fraud Detection* frequently include *Quantum Kernel* and other *Model*-category topics such as *Quantum Neural Network*. Generative model topics (e.g., *Quantum Circuit Born Machine* and *Quantum Generative Adversarial Network*) are also observed relatively often. Although fewer in number, *Privacy Enhancement*-category topics such as *Federated Learning* and *Quantum Differential Privacy* suggest an interest in privacy-conscious research.

Finally, Table VIII reveals that preprints in *Physics* primarily include *Quantum Kernel*, followed by

Model-category topics such as *Quantum Convolutional Neural Network*. The *Data*-category topic *Quantum Embedding* is also frequently noted.

IV. USE CASE IDENTIFICATION FROM SELECTED TOPICS

In this section, we identify and classify specific use cases from preprints associated with the candidate key topics selected in the previous section. By grouping similar use cases and examining their relative attention based on the number of preprints and citations, we aim to highlight the use case categories that may merit deeper investigation.

In the previous section, we identified four candidate key topics: *Medicine and Drug Discovery*, *Chemistry*, *Finance and Fraud Detection*, and *Physics*. Given our focus on potential industrial applications, we selected *Medicine*

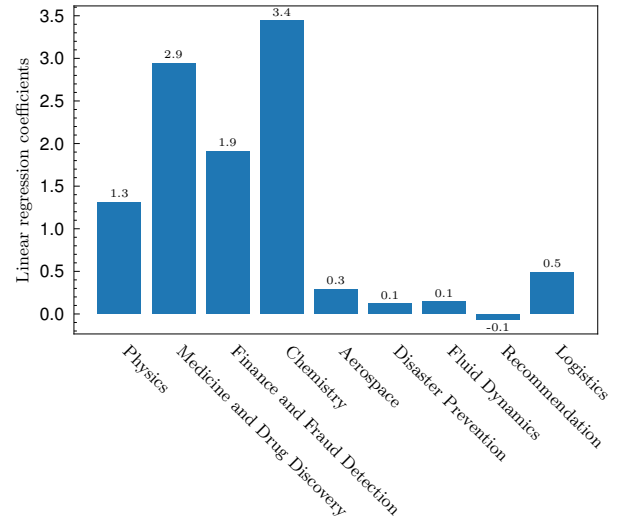
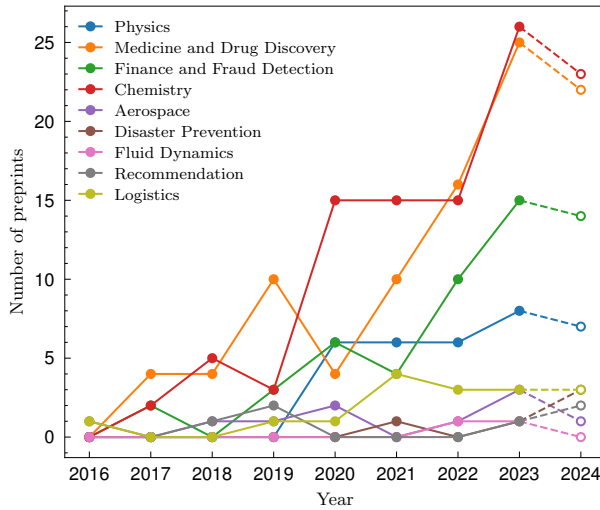


FIG. 4. Left panel: Annual number of preprints for each application topic from 2016 to 2024 (dashed lines indicate provisional data for 2024). Right panel: Linear regression coefficients of each application topic from 2016 to 2023.

TABLE V. Frequencies of technical topics in preprints associated with application topic *Medicine and Drug Discovery*.

Technical Topic	Preprints
Transfer Learning	10
Quantum Kernel	9
Quantum Neural Network	8
Quantum Data	5
Quantum Embedding	5
Federated Learning	3
Generalization	3
Tensor Network	3
Adversarial Attack	2
Expressivity	2
Graph	2
Quantum Circuit Born Machine	2
Quantum Generative Adversarial Network	2
Time Series	2
Circuit Encryption	1
Parametrized Quantum Circuit	1
Physics-informed Learning	1
Quantum Circuit Born Machine	1
Quantum Differential Privacy	1
Quantum Reservoir Computing	1
Quantum Speedup	1

and *Drug Discovery* and *Finance and Fraud Detection* as the two key topics for further detailed investigation. We chose not to focus on the topic *Physics* in our current analysis, as the identified cases primarily centered on high-energy physics and astrophysics, which are less relevant to our investigation of industrial applications. More-

TABLE VI. Frequencies of technical topics in preprints associated with application topic *Chemistry*.

Technical Topic	Preprints
Trainability	11
Quantum Neural Network	7
Quantum Embedding	7
Generalization	5
Quantum Data	5
Classical Surrogate/Shadow	4
Expressivity	4
Parametrized Quantum Circuit	3
Geometric Quantum Machine Learning	2
Quantum Autoencoder	2
Quantum Optimal Control	2
Quantum Reservoir Computing	2
Re-uploading Model	2
Tensor Network	2
Physics-informed Learning	1
Quantum Convolutional Neural Network	1
Quantum Kernel	1

over, because *Chemistry* is relatively mature in terms of research progress and existing demonstrations, we opted not to include it in our current analysis, preferring to investigate more emerging or less-established application areas.

To narrow the scope of our analysis, we initially focused on selected technical topics within *Medicine and Drug Discovery* and *Finance and Fraud Detection*. We further narrowed preprints covering the top four technical topics for each application topic (including ties) based on our analysis in the previous section. The resulting num-

TABLE VII. Frequencies of technical topics in preprints associated with application topic *Finance and Fraud Detection*.

Technical Topic	Preprints
Quantum Kernel	10
Quantum Neural Network	5
Quantum Circuit Born Machine	4
Quantum Generative Adversarial Network	4
Generalization	3
Quantum Data	3
Tensor Network	3
Federated Learning	2
Quantum Embedding	2
Re-uploading Model	2
Time Series	2
Trainability	2
Transfer Learning	2
Classical Surrogate/Shadow	1
Expressivity	1
Parametrized Quantum Circuit	1
Quantum Differential Privacy	1
Quantum Speedup	1
Reinforcement Learning	1

TABLE VIII. Frequencies of technical topics in preprints associated with application topic *Physics*.

Technical Topic	Preprints
Quantum Kernel	5
Quantum Convolutional Neural Network	4
Quantum Embedding	4
Parametrized Quantum Circuit	2
Classical Surrogate/Shadow	1
Geometric Quantum Machine Learning	1
Quantum Circuit Born Machine	1
Quantum Generative Adversarial Network	1
Quantum Autoencoder	1
Quantum Data	1
Quantum Speedup	1
Tensor Network	1

ber of preprints selected for further analysis is presented in Table IX, showing 29 preprints for *Medicine and Drug Discovery* and 20 preprints for *Finance and Fraud Detection*.

We manually extracted specific use cases from these selected preprints. We excluded use cases derived from experiments unrelated to the respective application topics, as well as those preprints that presented only theoretical discussions without experimental validation. We note that multiple use cases could be extracted from a single preprint.

Subsequently, we manually classified use cases into dis-

tinct use case categories. For each use case category, we counted the number of preprints and summed their number of citations.

The results of the extraction, classification, and aggregation of use cases are presented in Tables X and XI. Table X summarizes the use case classifications for *Medicine and Drug Discovery*, indicating that the categories *Drug Design and Evaluation* and *Medical Imaging Diagnosis* are relatively prominent based on the number of preprints and citations.

Table XI summarizes the use case classifications for *Finance and Fraud Detection*, showing that *Credit Card Fraud Detection* has the highest number of preprints. However, unlike *Medicine and Drug Discovery*, where a few categories clearly emerge as prominent, this category does not appear to be overwhelmingly dominant over the others.

V. EVALUATION OF EXPERIMENTAL VALIDATIONS AND QUANTUM ADVANTAGES

In the previous section, we extracted and categorized various use cases from selected application topics, identifying categories that have received notable attention based on publication volume and citations. In this section, we evaluate the experimental validations and reported outcomes related to these selected use cases. We particularly focus on how these studies assess potential quantum advantages in QML.

Specifically, we examine whether the studies explicitly evaluated the quantum advantage of QML algorithms compared to CML methods, using accuracy, speed, and energy consumption as the primary criteria [13]. Among the identified use case categories, we selected *Medical Imaging Diagnosis* (within *Medicine and Drug Discovery*) because of its prominence and broader applicability beyond medical domains. In addition, we chose *Credit Card Fraud Detection* (within *Finance and Fraud Detection*), as it included the largest number of preprints among the categories within the topic.

For each use case, we manually inspected the experimental validations. This process involved identifying which QML algorithms were used, which CML algorithms served as benchmarks, how many qubits were utilized, and whether the experiments ran on quantum hardware or simulators. We also assessed how these studies addressed quantum advantage by comparing QML performance with CML methods in terms of accuracy, speed, and energy consumption. Furthermore, we noted explicit references to quantum advantage, future research challenges, and practical implications.

Our survey indicates that none of the reviewed preprints explicitly discussed energy consumption. Regarding speed, only one study provided measured computational times, and even in that instance, no direct comparison with CML methods was made. Nonetheless, another study offered estimated computational times and

TABLE IX. Number of preprints per technical topic and total number of unique preprints (accounting for overlaps). Preprints listed here were used as the source for use case extraction.

Key Topic	Technical Topics and Number of Preprints	Unique Preprints
Medicine and Drug Discovery	Model category Transfer Learning: 10, Quantum Kernel: 9, Quantum Neural Network: 8	29
	Data category Quantum Data: 5, Quantum Embedding: 5	
Finance and Fraud Detection	Model category Quantum Kernel: 10, Quantum Neural Network: 5, Quantum Circuit Born Machine: 4, Quantum Generative Adversarial Network: 4	20

TABLE X. Use case categories from *Medicine and Drug Discovery*, with numbers of preprints, citations, and use cases for each category.

Use Case Category	Preprints	Citations	Use Case(s)
Drug Design and Evaluation	7	133	Drug Candidates Discovery [34, 56] Predicting IC50 Values for Anti-cancer Drugs [35] Toxicity Screening on Molecules [45, 57] Predicting the Binding Affinity of Drug Compounds [58] Peptide Classification by Hemolytic Activity [59]
Medical Imaging Diagnosis	7	120	Diabetic Retinopathy Detection from Retinal Fundus Images [60] Pneumonia Detection from Chest X-ray Images [60, 61] Classification of Hand and Breast MRI Scans [62] Cancer Detection from Digital Pathology Images [63] Detection of COVID-19 from CT Scans of the Lung [61] CT Scans Classification into Bladder, Femur, and Heart [61] Classification of Breast Cancer Images [64] Hepatic Steatosis Diagnostics from Liver Biopsy Images [65] Dementia Staging from MRI Scans [66]
Clinical Record Analysis	5	36	Patient Persistence Prediction on Biologic Therapies [67] Classification on a Breast Cancer Biomarkers Dataset [68–71]
Genomic and Omics Analysis	2	0	Glioma Classification from Molecular and Clinical Attributes [71] Classification of Breast Cancer using Omics Data [72]
Physiological Data Analysis	1	3	Neurological Stress Level Estimation from Multi-Model Sensing [73] Emotion Elicitation Estimation from EEG [73] Motor Imagery Estimation from EEG [73] Detection of Responses from EEG for System Errors [73] ECoG Detection on Face-vs-House Discrimination Task [73] Finger Gesture Identification from EMG [73]

discussed the implications in detail (see Subsection V B).

The following subsections provide detailed insights, starting with *Medical Imaging Diagnosis*, and then moving on to *Credit Card Fraud Detection*.

A. Survey Results: Medical Imaging Diagnosis

We present our survey findings on the experimental validations of ten use cases in the category *Medical Imag-*

ing Diagnosis. Table XII summarizes key information, including the year of publication on arXiv, number of qubits, type of quantum hardware, QML model type, and accuracy advantage of the QML model over CML models. Additional details are provided in the following.

First, we outline the validation setups observed across these studies. Quantum neural networks were commonly employed as QML models, reflecting their popularity in the field. Hybrid neural network approaches, using classical networks to extract features from image data and

TABLE XI. Use case categories from *Finance and Fraud Detection*, with numbers of preprints, citations, and use cases for each category.

Use Case Category	Preprints	Citations	Use Case(s)
Credit Card Fraud Detection	4	68	Credit Card Fraud Detection [39, 40, 74, 75]
Portfolio Optimization	3	84	Scenarios Construction for Portfolio Optimization [76] Stock Portfolio Optimization [77] Generating Probability Distributions of Portfolio Returns [78]
Financial Time Series Generation	3	55	Learning Distribution of Correlated Currency Pairs [79] Reproduction of Financial Model Distributions [80] Options Pricing using Financial Dataset [81]
Credit Risk and Default Prediction	3	39	Loan Default Prediction [82, 83] Personal Credit Risks Classification [84]
Marketing and Churn Prediction	2	39	Banking Customer Churn Prediction [82] Subscription Prediction of a Term Deposit [84]
Banking Transaction Fraud Detection	2	14	Banking Transaction Fraud Detection [84, 85]
Asset Price Prediction	2	6	Housing Price Prediction [86] UK Nickel Price Prediction [74]
Derivative Hedging Optimization	1	18	Derivative Hedging Optimization [38]
Crypto Asset Fraud Detection	1	4	Phishing Detection in Ethereum Network [87]

QML models to enhance feature discrimination, were also prevalent. Most studies evaluated accuracy metrics by comparing QML and CML models with the same extracted features. The number of qubits ranged from 4 to 10, and the majority of experiments were conducted on quantum simulators. None of the surveyed papers reported measurements of computational time or energy consumption.

Next, we review the outcomes of these experimental validations. Although QML models demonstrated practical accuracy across several use cases, their improvements over CML models were generally modest, typically within a few percentage points. A notable exception was observed in a study on dementia staging from Magnetic Resonance Imaging (MRI) scans [66], where the QML approach significantly outperformed the CML counterpart: 96.1% versus 78.8% accuracy. The authors attribute this result specifically to the QML model’s ability to detect subtle changes associated with early-stage cognitive decline. They suggest that this advantage arise because neuroimaging data are inherently complex and high-dimensional, a setting in which QML models are hypothesized to better capture subtle patterns compared to CML models. Additionally, some studies indicated stable model accuracy even with smaller datasets [63, 65], suggesting that QML may offer advantages in data-limited medical settings. On the other hand, one study found a substantial drop in accuracy when noise was introduced, particularly if the noise was present during training process [60].

Finally, we discuss the future challenges highlighted in these studies. The limited number of qubits emerges as a common concern [60, 61], as large imaging datasets often require higher-dimensional feature spaces. Al-

though feature dimension reduction via classical neural networks offers a temporary workaround, expanding the available qubits may allow more direct incorporation of high-dimensional features into QML models. Such advancements could further improve performance in medical imaging applications.

B. Survey Results: Credit Card Fraud Detection

We present our survey findings on the experimental validations of four use cases in the category *Credit Card Fraud Detection*. Table XIII provides an overview of these results.

First, we describe the validation setups observed across these studies. Quantum Kernel Support Vector Machines (QK-SVM) emerged as the predominant QML models. In QK-SVM, data points are mapped into quantum states via feature maps, and their inner products, computed on a quantum computer, form a quantum kernel. This quantum kernel is then utilized within a classical support vector machine (SVM) to perform the classification. Each study compared accuracy with CML methods using identical feature sets, typically reduced via Principal Component Analysis (PCA) or feature selection to match the specified number of qubits. The number of qubits ranged from 2 to 20, and most experiments were conducted on quantum simulators, with some incorporating noise models [40, 74]. Although one study presented estimated computational times [39] and another reported actual measurements [75], neither directly compared these results with CML counterparts. None of the surveyed papers addressed energy consumption.

TABLE XII. Summary of survey results for each use case in *Medical Imaging Diagnosis*. Quantum neural network-based models are denoted as **QNN**, quantum kernel support vector machine-based models as **QK-SVM**, and hybrid models combining QML and CML models as **Hybrid**. Accuracy advantage categories are defined based on the percentage difference between best QML and CML results reported in each study as follows: **Superior** (greater than +5%), **Slightly Better** (+1% to +5%), **Comparable** (between -1% and +1%), **Slightly Worse** (-5% to -1%), and **Inferior** (less than -5%). Specific metrics used, such as the Area Under the ROC Curve (AUC) and the accuracy are indicated in parentheses, e.g., Superior (AUC).

Use case	Year (arXiv)	Used Qubits	Hardware	Model	Accuracy Advantage
Pneumonia Detection from Chest X-ray Images [60]	2021	5–9	Superconducting Simulator	QNN	Comparable (AUC) Slightly Better (Accuracy)
Diabetic Retinopathy Detection from Retinal Fundus Images [60]	2021	5–9	Superconducting Simulator	QNN	Comparable (AUC, Accuracy)
Classification of Hand and Breast MRI Scans [62]	2022	10	Superconducting	QNN	Not Compared
Cancer Detection from Digital Pathology Images [63]	2023	4–7	Simulator	Hybrid QNN	Slightly Worse (AUC, Accuracy)
Detection of COVID-19 from CT Scans of the Lung [61]	2023	4–9	Simulator	Hybrid QNN	Comparable (AUC, Accuracy)
Pneumonia Detection from Chest X-ray Images [61]	2023	4–9	Simulator	Hybrid QNN	Comparable (AUC, Accuracy)
CT Scans Classification into Bladder, Femur, and Heart [61]	2023	4–9	Simulator	Hybrid QNN	Slightly Worse (AUC) Inferior (Accuracy)
Classification of Breast Cancer Images [64]	2023	8–10	Simulator	Hybrid QNN	Slightly Better (F1 Score, Recall) Comparable (Precision)
Hepatic Steatosis Diagnostics from Liver Biopsy Images [65]	2023	5	Simulator	Hybrid QNN	Slightly Better (Accuracy)
Dementia Staging from MRI Scans [66]	2024	8	Simulator	Hybrid QK-SVM	Superior (Accuracy)

TABLE XIII. Summary of survey results for each use case in *Credit Card Fraud Detection*. Columns are identical to those described in Table XII.

Use case	Year (arXiv)	Used Qubits	Hardware	Model	Accuracy Advantage
Credit Card Fraud Detection [39]	2022	2–20	Simulator	QK-SVM	Superior (Average Precision)
Credit Card Fraud Detection [40]	2022	3–7	Simulator	QK-SVM	Comparable (Accuracy) Slightly Worse (AUC)
Credit Card Fraud Detection [74]	2023	4–12	Trapped-ion Simulator	QK-SVM	Comparable (Accuracy)
Credit Card Fraud Detection [75]	2024	7	Simulator	QNN	Comparable (Precision) Inferior (Accuracy, Recall, F1 Score)

Next, we examine the validation outcomes. Notably, one study [39] found that its QK-SVM model achieved an average precision of around 70%, an improvement of approximately 15 percentage points over the CML counterpart in an unsupervised scenario. In another study [40], the authors observed differences in the emphasized features and decision patterns between decision-tree-based CML models and QK-SVM. They also demonstrated that combining QK-SVM with a meta-classifier to adjudicate conflicting predictions improved accuracy. However, introducing noise models degraded accuracy, underscoring the sensitivity of QML approaches to noisy conditions [40, 74]. In particular, the study [74] examined differences between noisy and ideal quantum kernels at varying error rates, discussing how these differences impact model accuracy.

Regarding future challenges, one study [39] emphasizes the need for additional qubits and deeper circuits to achieve further quantum advantage, as well as the importance of developing unsupervised methods for detecting emerging fraud patterns. From an economic perspective, the authors note that a 15% enhancement in fraud detection accuracy could translate into substantial savings, estimating that in the United Kingdom alone, this improvement could save approximately £150 million annually.

Finally, computational time remains a significant concern. The same study [39] discusses the considerable runtime required for quantum kernel calculations on large datasets. For example, estimates suggest that, even with a superconducting quantum processor capable of one microsecond per shot, achieving high-quality estimation (500 data points and 10^6 shots) could take up to 28 hours, extending to 16 weeks for datasets with 10^5 data points and 1000 shots. To mitigate these overheads, the authors propose kernel evaluation through randomized measurements [88], indicating that this approach could reduce a 16-week computation to about one minute.

VI. SUGGESTED VALIDATION APPROACHES FOR THE SURVEYED USE CASES

In the previous section, we surveyed the experimental validations, potential quantum advantages, and future challenges observed in the use case categories *Medical Imaging Diagnosis* and *Credit Card Fraud Detection*. Building on these findings, this section offers our perspective on suggested validation approaches and research directions for these specific applications.

A. Suggestion: Medical Imaging Diagnosis

The surveyed studies used relatively small numbers of qubits. At the time of writing, quantum computers capable of handling around 100 qubits are accessible. This makes it feasible to greatly expand both the number of

qubits and the dimensionality of input features. Future validations should focus on determining whether these expansions lead to substantial accuracy gains over CML models.

However, increasing the number of qubits and the circuit depth raises concerns related to the Barren Plateau (BP) phenomenon [89, 90], in which gradients vanish and hinder the effective training of quantum neural networks. Hence, strategies to mitigate the BP problem become essential. Furthermore, when validations surpass the capacity of simulators, quantum hardware must be used. Such hardware inevitably introduces noise, a factor shown in our survey to degrade model accuracy. Consequently, employing effective error mitigation techniques is critical. These two considerations, which are rarely addressed in the surveyed studies, are key to scaling up quantum experiments.

Moreover, maintaining model accuracy is especially critical when only small sample sizes are available, a common scenario in medical contexts. Therefore, future research should pay closer attention to this issue. Although theoretical guarantees for QML's generalization capabilities with small datasets have been explored [91], these findings remain debated [92] and are not yet conclusively established.

Regarding quantum advantage in medical imaging diagnosis, one particularly promising example from our survey is the dementia staging task based on MRI scans. In that study, a QML model significantly outperformed CML approaches in detecting subtle differences associated with early-stage cognitive decline [66]. This improvement can be attributed to the inherent complexity and high dimensionality of neuroimaging data, which QML models may handle more effectively than CML methods by capturing discriminative features more accurately. Such findings suggest that pursuing validations on challenging problems, where CML models often struggle to achieve high accuracy, could be especially worthwhile. Hence, identifying and examining similar datasets or tasks should be a priority.

B. Suggestion: Credit Card Fraud Detection

One of the surveyed studies indicates potential quantum advantages in unsupervised anomaly detection [39], aligning with the practical need to identify novel fraud types. Future research should explore this potential more extensively. According to our survey results, a key question for validation is whether such advantages persist or grow when scaling beyond 20 qubits. In particular, increasing the number of qubits would allow direct use of higher-dimensional, real-world data without resorting to dimensionality reduction methods like PCA, thereby enhancing practical applicability.

However, the open dataset [93] used in our surveyed studies contains only 28 features, which inherently limits the ability to increase the number of qubits if the same

feature map is applied. Since datasets specific to credit card fraud are often unavailable publicly, future validations should explore alternative open datasets, although the options are limited, such as the IEEE-CIS Fraud Detection dataset [94], or proprietary datasets maintained by financial institutions.

Additionally, scaling up the number of qubits may give rise to the phenomenon of exponential concentration [95–97], in which kernel values converge exponentially toward a single value and can degrade model performance. While a previous study with 20 qubits employed noiseless statevector simulators [39], future work using noisy quantum simulators or real quantum hardware must carefully assess the impact of even slight measurement inaccuracies on model performance. If performance drops occur, addressing exponential concentration and applying appropriate error mitigation strategies will be crucial.

For example, a recent paper published after our survey period (end of September 2024) reported achieving classical-level accuracy in QK-SVM using 156 qubits [98]. The authors introduced a novel error mitigation method, called “Bit Flip Tolerance,” specifically designed to address exponential concentration.

As our survey suggests, computational time of kernel evaluation remains a major bottleneck in practical QK-SVM applications. Consequently, methods to reduce kernel evaluation time have been actively explored. Specifically, recent studies [99, 100] experimentally evaluated two techniques, randomized measurements and variable subsampling, as well as their combination. Using the same dataset [93] and problem settings as those in Reference [39], these experiments demonstrated that computational time for kernel evaluation could be reduced to linear complexity with respect to dataset size. Because conventional kernel evaluation typically scales quadratically with dataset size, the computational advantage becomes increasingly pronounced as datasets grow larger. As a future direction, developing scalable methods that reduce computational time while maintaining acceptable model performance will be crucial to enable QK-SVM for real-world applications.

VII. SUMMARY AND FUTURE DIRECTIONS

We have systematically investigated the current landscape of QML research to bridge the gap between theoretical/industrial promise and the lack of clear use cases demonstrating advantages over CML. We briefly summarize the major findings from our survey. We then provide a set of future research directions in light of these findings.

A. Summary and Conclusion

To address the first research question posed in the Introduction—which areas and use cases are promising for

QML in near-term applications—we systematically analyzed 2539 QML-related preprints from arXiv. Based on publication and citation trends, we identified four primary application areas, selecting *Medicine and Drug Discovery*, and *Finance and Fraud Detection* as particularly prominent. Within these fields, *Medical Imaging Diagnosis* and *Credit Card Fraud Detection* emerged as representative high-potential use cases. Our findings thus highlight key domains and use cases aligned with near-term opportunities for QML.

To address the second research question—what advantages are expected and what challenges exist regarding the practical realization of QML—we examined empirical validations and future challenges for two representative use cases: *Medical Imaging Diagnosis* and *Credit Card Fraud Detection*. For *Medical Imaging Diagnosis*, QML models demonstrated advantages in accurately identifying subtle patterns in neuroimaging data and stable performance with smaller datasets. For *Credit Card Fraud Detection*, QML models outperformed CML models particularly in unsupervised anomaly detection tasks. Despite these promising advantages, both cases highlight shared challenges, including increasing the number of qubits and input feature dimensions, addressing issues like Barren Plateau and exponential concentration phenomena, employing error mitigation methods, and reducing computational time.

Finally, we provide overall conclusions from our analysis. Through this survey, we identified several promising areas and use cases, and further clarified the current status, challenges, and future directions regarding the expected quantum advantages of QML. Our study has inherent limitations, as we employed statistical methods to initially narrow down relevant studies and subsequently selected only a subset of topics deemed particularly significant, which may have resulted in not fully covering other potentially important areas of investigation. Nevertheless, even within the scope examined here, we have gained meaningful insights into potential future directions for QML. Therefore, we believe our analysis has partially addressed the two research questions. Investigating additional important topics identified but not included in this detailed survey, such as *Chemistry*, could offer a more comprehensive understanding of the overall landscape of QML.

B. Future Directions

While Section VI outlined future directions for validating individual use cases, here we present broader future directions for the field of QML as a whole. We structured our assessment of quantum advantages according to the three criteria of accuracy, speed, and energy consumption as proposed in prior work [13]. Accordingly, based on this framework of three criteria, we provide some perspectives on future research directions.

Accuracy In most of the preprints included in our survey, validation was conducted from the perspective of accuracy, typically under fair conditions such as utilizing data of identical dimensionality suitable for the limited resources of quantum simulators and NISQ computers. QML models demonstrated slight advantages over CML counterparts in some cases under these conditions, but definitive quantum advantage remains elusive. Notably, some specific cases indicate scenarios where CML struggled but quantum methods excelled. These scenarios suggest a fruitful area for targeted investigation. Additionally, in medical domains where dataset sizes are often limited, QML’s potential capability to achieve high accuracy with fewer samples remains an exciting area of future exploration.

Speed NISQ-era algorithms have demonstrated quantum speedup from the perspective of computational complexity for specific problem settings [11, 12], but these advantages have not been established in general, unlike fully FTQC algorithms. Practical measurement of computational times is rarely addressed in our surveyed preprints, suggesting a lack of attention to this aspect. Possible reasons include practical issues such as job waiting times and communication delays between quantum and conventional (classical) computers. Currently, platforms are being developed that co-locate quantum computers with High-Performance Computing (HPC) systems, connected via low-latency, high-bandwidth networks with coordinated scheduling [101]. Such platforms will reduce waiting and communication times, potentially enabling meaningful speed comparisons and assessments.

Energy The potential quantum advantage in energy consumption deserves particular attention. As CML models, especially foundation models such as Large Language Models (LLMs) and image generation models, become increasingly resource-intensive [102], significant reductions in energy usage through QML could represent a major advantage, even if QML algorithms only match CML performance in terms of accuracy. However, to the best of our knowledge, there are no empirical demonstrations confirming such energy-related advantages of QML, nor is energy consumption measured in any surveyed studies. Establishing the energy efficiency of QML algorithms remains a critical and largely unexplored area. As noted by Reference [103], the lack of an established methodology to evaluate quantum computing’s energy advantages has led to initiatives such as the interdisciplinary “Quantum Energy Initiative (QEI)” connecting quantum thermodynamics, quantum information science, quantum physics and engineering. Although various studies have emerged following QEI’s proposal, realizing empirical demonstrations of energy advantages will require dedicated quantum computing platforms designed for energy measurements, whose development we anticipate with considerable interest.

DATA AVAILABILITY

All datasets necessary to reproduce the results presented in this paper, including those described in the Appendix, have been deposited in Zenodo and are publicly available. These include:

- the metadata of 2539 arXiv preprints and their associated citation counts,
- processed datasets used for topic extraction and trend analysis, and
- manually curated files derived from expert Inspection-Based topic extraction and use case identification.

The datasets are accessible at: <http://dx.doi.org/10.5281/zenodo.17861466>.

CODE AVAILABILITY

All code required to reproduce the analyses and visualizations in this study, including the four Jupyter notebooks corresponding to each stage of the workflow, is publicly available in the same Zenodo repository. A Dockerfile and docker-compose.yml file are also provided to facilitate reproducibility of the execution environment. The code and environment files are accessible at: <http://dx.doi.org/10.5281/zenodo.17861466>.

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Appendix A: List of Statistically Extracted Topics

This appendix provides the complete list of “statistically extracted topics” described in Section II. The topics presented here were determined by manual labeling based on highly-appeared keywords within preprint clusters obtained via t-SNE and KDE. The top-ranked keywords were identified based on the number of preprints in which each keyword appeared in the abstract texts. The full list is presented in Table XIV, where only the top three ranked keywords per topic are shown due to space constraints. The full keyword sets, together with additional details and the code used for keyword extraction, are available in the public repository referenced in the availability statements.

TABLE XIV. List of statistically extracted topics, central density of each cluster computed by KDE, number of preprints contained in each cluster, and highly-appeared keywords of each cluster. Topics are sorted by their central density. Keywords are pre-processed, and only the top three keywords for each topic are listed.

Extracted Topic	Central Density	Preprints	Highly-appeared Keywords
Quantum Kernel	2.91×10^{-4}	66	“kernel”, “featur”, “classif”
Quantum Reservoir Computing	2.20×10^{-4}	47	“reservoir”, “dynam”, “task”
Phase Transition*	2.12×10^{-4}	33	“phase”, “transit”, “order”
Tensor Network	2.06×10^{-4}	43	“tensor”, “physic”, “effici”
Error Correction*	2.05×10^{-4}	37	“error”, “code”, “correct”
Parametrized Quantum Circuit	2.03×10^{-4}	36	“pqc”, “parameter”, “howev”
Quantum Neural Network	1.95×10^{-4}	29	“neural”, “qnn”, “advantag”
Entanglement*	1.92×10^{-4}	37	“entangl”, “measur”, “separ”
Implementation**	1.86×10^{-4}	68	“gate”, “qubit”, “implement”
Photonic Quantum Computer**	1.84×10^{-4}	58	“photon”, “optic”, “implement”
Performance (category rather than single topic)	1.83×10^{-4}	47	“qml”, “ha”, “potenti”
Image Classification***	1.79×10^{-4}	42	“imag”, “classif”, “neural”
Geometric Quantum Machine Learning	1.74×10^{-4}	36	“symmetri”, “equivari”, “group”
Adversarial Attack	1.69×10^{-4}	51	“adversari”, “robust”, “attack”
Trainability	1.69×10^{-4}	54	“gradient”, “paramet”, “one”
Quantum Convolutional Neural Network	1.67×10^{-4}	28	“neural”, “convolut”, “qcnv”
Quantum Tomography*	1.67×10^{-4}	26	“reconstruct”, “tomographi”, “measur”
Tuning Quantum Computer*	1.61×10^{-4}	32	“dot”, “devic”, “tune”
Reinforcement Learning	1.60×10^{-4}	38	“reinforc”, “ha”, “recent”
Quantum Speedup	1.60×10^{-4}	22	“epsilon”, “time”, “improv”
Quantum Annealing****	1.59×10^{-4}	43	“anneal”, “ha”, “sampl”
Quantum Speedup (HHL Algorithm)	1.59×10^{-4}	36	“equat”, “solv”, “also”
Chemistry	1.55×10^{-4}	44	“energi”, “molecular”, “molecul”
Federated Learning	1.53×10^{-4}	2	“advanc”, “commun”, “compar”
Graph	1.53×10^{-4}	22	“graph”, “neural”, “paper”
Generalization / Error Mitigation*	1.52×10^{-4}	32	“nois”, “devic”, “error”
Parametrized Quantum Circuit	1.42×10^{-4}	18	“variat”, “vqc”, “neural”
Restricted Boltzmann Machine****	1.40×10^{-4}	15	“boltzmann”, “restrict”, “rbm”
Quantum Neural Network	1.39×10^{-4}	10	“perceptron”, “paper”, “neural”
Quantum Speedup (Clustering)	1.31×10^{-4}	17	“cluster”, “mean”, “implement”
Chemistry	1.30×10^{-4}	55	“mani”, “bodi”, “variat”
Chemistry	1.19×10^{-4}	4	“dynam”, “hamiltonian”, “spin”
QUBO Feature Selection****	1.19×10^{-4}	22	“quadrat”, “binari”, “qubo”
Finance and Fraud Detection	1.03×10^{-4}	14	“anomali”, “detect”, “task”
Quantum Autoencoder	9.70×10^{-5}	14	“autoencod”, “compress”, “dimension”
Bell’s Inequality**	9.17×10^{-5}	14	“bell”, “causal”, “also”
Magnetic Material*	8.80×10^{-5}	10	“magnet”, “here”, “diamond”
Quantum Key Distribution*	8.51×10^{-5}	11	“distribut”, “key”, “qkd”
Quantum Information***	6.45×10^{-5}	5	“entropi”, “inform”, “neumann”
Graphical Method**	6.13×10^{-5}	9	“calculu”, “diagram”, “zx”
Review/Lecture**	4.17×10^{-5}	5	“into”, “cover”, “inform”
Quantum Speedup (Topological Data Analysis)	4.07×10^{-5}	5	“analysi”, “betti”, “number”
Logistics	2.53×10^{-5}	3	“ansatz”, “approxim”, “citi”
Quantum Probability**	1.68×10^{-5}	2	“address”, “ani”, “calcul”
Nonclassicality*	1.67×10^{-5}	2	“artifici”, “experiment”, “here”

Topics marked with asterisks (*, **, ***, ****) were excluded from statistically extracted topics for the following reasons:

*Relevant to the utilization of CML.

**Not directly relevant to QML.

***Too general as specific topics.

****Relevant to quantum annealing approaches, outside survey scope.

Appendix B: Detailed Descriptions of Technical Topics

This appendix provides detailed descriptions of technical topics identified in Section II, organized into six categories: *Model*, *Data*, *Performance*, *Quantum-Classical Synergy*, *Privacy Enhancement*, and *Quantum Speedup*. The category *Model* is further divided into two parts, Variational methods and Non-variational methods. Technical topics in each category and their descriptions are summarized in Tables XV–XXI below.

Appendix C: Search Keywords for Technical Topics Used in Trend Analysis

This appendix provides the search keywords and exclusion keywords used for identifying technical topics in the keyword-based trend analysis described in Section III. The keywords are organized into six technical categories: *Model*, *Data*, *Performance*, *Quantum-Classical Synergy*, *Privacy Enhancement*, and *Quantum Speedup*. The category *Model* is further divided into two parts, Variational methods and Non-variational methods. Tables XXII–XXVIII summarize the keywords defined for each category. It is noted that words enclosed in parentheses indicate keyword pairs that are counted as hits only when all enclosed words co-occur within the same abstract text.

TABLE XV. List of technical topics in the category *Model* (Variational methods), descriptions for each topic, and number of preprints contained in each topic. The “Preprints” column shows the frequency of each topic within the 245 highly-cited preprints subjected to expert inspection.

Technical Topic	Description	Preprints
Parametrized Quantum Circuit (PQC)	QML method using quantum circuits with parameters as models (Also known as “Quantum Circuit Learning”)	22
Geometric Quantum Machine Learning	A type of QNN using circuits that reflect the symmetry and geometric structure of the data	13
Quantum Neural Network (QNN)	QML method using PQCs with layered structure	11
Quantum Convolutional Neural Network	A type of QNN using circuits with convolutional layers and pooling layers	10
Transfer Learning	A type of QNN combining layers of classical neural networks with layers of PQC	9
Reinforcement Learning	Reinforcement learning method using PQCs	8
Quantum Generative Adversarial Network	Generative adversarial network based method using QNNs in either or both generator and discriminator models.	6
Quantum Circuit Born Machine	Generative QML method using PQCs	5
Quantum Recurrent Neural Network	A type of QNN using circuits with recurrent layers	2
Re-uploading Model	A type of QNN using alternating data embedding and QNN layers	2
Quantum Optimal Control	QML method using pulses for gate operations as model parameters	2
Quantum Autoencoder	A type of QNN compressing information into fewer qubits	1
State Diagonalization	Method of diagonalizing quantum states using PQCs	1
QAOA-clustering	Clustering method using Quantum Approximate Optimization Algorithm (QAOA) for combinatorial optimization	1

TABLE XVI. List of technical topics in the category *Model* (Non-variational methods), descriptions for each topic, and number of preprints contained in each topic. The “Preprints” column shows the frequency of each topic within the 245 highly-cited preprints subjected to expert inspection.

Technical Topic	Description	Preprints
Quantum Kernel	QML method using kernels defined as inner products of data-encoded quantum states	21
Quantum Reservoir Computing	Reservoir computing method using quantum circuits as physical reservoirs	7
Imaginary Time Evolution	Method that accelerates CML by replacing Markov chain Monte Carlo sampling in the learning process with sampling from ground states generated via Imaginary Time Evolution.	2

TABLE XVII. List of technical topics in the category *Data*, descriptions for each topic, and number of preprints contained in each topic. The “Preprints” column shows the frequency of each topic within the 245 highly-cited preprints subjected to expert inspection.

Technical Topic	Description	Preprints
Quantum Embedding	Topics on how to embed classical data into quantum circuits	5
Quantum Data	Topics on data obtained as quantum states	2
Time Series	Topics in QML with time series data	2
Graph	Topics in QML with graph-structured data	2
Physics-informed Learning	Topics in QML that incorporate physical laws as well as data in learning	1

TABLE XVIII. List of technical topics in the category *Performance*, descriptions for each topic, and number of preprints contained in each topic. The “Preprints” column shows the frequency of each topic within the 245 highly-cited preprints subjected to expert inspection.

Technical Topic	Description	Preprints
Trainability	Topics on the trainability and the Barren Plateau (vanishing gradient) problem of QML models	24
Generalization	Topics on the generalization performance and the overfitting problem of QML models	13
Adversarial Attack	Topics on the resistance of QML models to adversarial attacks	3
Expressivity	Topics on the expressive power of QML models	1

TABLE XIX. List of technical topics in the category *Quantum-Classical Synergy*, descriptions for each topic, and number of preprints contained in each topic. The “Preprints” column shows the frequency of each topic within the 245 highly-cited preprints subjected to expert inspection.

Technical Topic	Description	Preprints
Classical Surrogate/Shadow	Topics on techniques enabling inference on conventional computers by replacing QML models with CML counterparts.	6
Tensor Network	Topics on techniques accelerating QML model training using initial parameters obtained through pre-training tensor networks on conventional computers.	4
Training Classical Model	Topics on techniques for efficient training by embedding classical neural network models into quantum neural networks with fewer parameters.	4

TABLE XX. List of technical topics in the category *Privacy Enhancement*, descriptions for each topic, and number of preprints contained in each topic. The “Preprints” column shows the frequency of each topic within the 245 highly-cited preprints subjected to expert inspection.

Technical Topic	Description	Preprints
Federated Learning	Topics on implementing federated learning, training a single model while keeping sensitive data decentralized, through QML.	5
Quantum Differential Privacy	Topics on implementing differential privacy, a protection technique for sensitive classical data, with quantum information	2
Circuit Encryption	Topic on technology for encrypting quantum circuits to prevent information leakage from data-embedded quantum circuits	1

TABLE XXI. List of technical topics in the category *Quantum Speedup*, descriptions for each topic, and number of preprints contained in each topic. The “Preprints” column shows the frequency of each topic within the 245 highly-cited preprints subjected to expert inspection.

Technical Topic	Description	Preprints
Quantum Speedup	Topics on quantum acceleration of some processes in CML algorithms (regression, principal component analysis, support vector machine, clustering, topological data analysis, etc.) by using quantum algorithms for FTQC such as the Harrow-Hassidim-Lloyd algorithm, Grover algorithm, quantum phase estimation, etc.	15

TABLE XXII. Keywords and exclusion keywords for technical topics of the category *Model* (Variational methods).

Technical Topic	Keyword(s)	Exclusion Keyword(s)
Parametrized Quantum Circuit	“parametrized quantum circuit,” “parametrised quantum circuit,” “pqc,” “quantum circuit learning,” “qcl,” “variational quantum classification,” “quantum variational classification”	None
Quantum Neural Network	“quantum neural,” “qnn,” “quantum perceptron”	None
Quantum Convolutional Neural Network	“quantum convolutional,” “qcnn,” “quanvolutional”	None
Geometric Quantum Machine Learning	“geometric quantum machine learning,” “gqml,” “equivariant,” “group invariant”	None
Quantum Recurrent Neural Network	“quantum recurrent,” “qrnn”	None
Quantum Generative Adversarial Network	“quantum generative adversarial,” “qgan,” “q gan”	None
Quantum Autoencoder	“quantum autoencoder”	None
Re-uploading Model	“re uploading,” “reuploading”	None
Transfer Learning	“quantum transfer learning,” “hybrid transfer learning,” “hybrid quantum neural,” “quantum classical neural”	None
Quantum Circuit Born Machine	“born machine,” “qcbm”	None
Reinforcement Learning	“quantum reinforcement,” “qrl”	None
State diagonalization	“state diagonalization”	None
QAOA-clustering	(“quantum approximate optimization algorithm,” “clustering”), (“qaoa,” “clustering”)	None
Quantum Optimal Control	“quantum optimal control,” “variational quantum pulse”	None

TABLE XXIII. Keywords and exclusion keywords for technical topics of the category *Model* (Non-variational methods).

Technical Topic	Keyword(s)	Exclusion Keyword(s)
Quantum Kernel	“quantum kernel,” “quantum support vector,” “qsvm,” “quantum feature map”	None
Quantum Reservoir Computing	“quantum reservoir,” “quantum extreme learning machine”	None
Imaginary Time Evolution	“imaginary time evolution”	None

TABLE XXIV. Keywords and exclusion keywords for technical topics of the category *Data*.

Technical Topic	Keyword(s)	Exclusion Keyword(s)
Quantum Embedding	“quantum embedding,” “quantum embedded,” “loading classical,” “load classical,” “loads classical”, (“encode”, “data”)	None
Quantum Data	“quantum data,” “entangled data”	None
Time Series	“timeseries,” “time series”	None
Graph	“graph structured,” “graph data”	None
Physics-informed Learning	“physics informed neural”	None

TABLE XXV. Keywords and exclusion keywords for technical topics of the category *Performance*.

Technical Topic	Keyword(s)	Exclusion Keyword(s)
Expressivity	“expressivity”	None
Generalization	“generalization,” “generalisation,” “overfitting,” “over fitting,” “quantum fisher information”	None
Trainability	“trainability,” “barren plateau,” (“gradient,” “vanish”)	None
Adversarial Attack	“adversarial attack,” “adversarial example”	None

TABLE XXVI. Keywords and exclusion keywords for technical topics of the category *Quantum-Classical Synergy*.

Technical Topic	Keyword(s)	Exclusion Keyword(s)
Classical Surrogate/Shadow	“classical surrogate,” “classical shadow,” “shadow model,” “pauli shadow,” “classical approx”	None
Tensor Network	“quantum tensor network,” (“tensor network,” “circuit”)	None
Training Classical Model	(“quantum train,” “qt”)	None

TABLE XXVII. Keywords and exclusion keywords for technical topics of the category *Privacy Enhancement*.

Technical Topic	Keyword(s)	Exclusion Keyword(s)
Circuit Encryption	“encrypt”	“quantum key distribution,” “qkd,” “random generator”
Federated Learning	“federated”	None
Quantum Differential Privacy	“differential privacy”	None

TABLE XXVIII. Keywords and exclusion keywords for technical topics of the category *Quantum Speedup*.

Technical Topic	Keyword(s)	Exclusion Keyword(s)
Quantum Speedup	“quantum speedup,” “quantum speed up,” “harrow hassidim lloyd,” “hhl,” “quantum k means,” “quantum nearest neighbor”	None

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